Nada Mohamed Sayed

Software Engineer

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Egypt

Profile

Software Engineering with a diverse technical background in Software Development, Backend Development, Data Analysis, Data Science, and Applications Integration. Experienced in building practical projects using Python, Java, C++, SQL, and Spring Boot, with knowledge of Microservices, REST APIs, Docker, Apache Kafka, and WSO2. Skilled in problem-solving, analytical thinking, and transforming data into meaningful insights. Eager to contribute to real-world projects and continuously grow as a versatile technology professional.

Education

Bachelor's Degree in Computer and Information Science, Major in Software Engineering, *Assiut University* 2022 – 2026
GPA: 3.94 (out of 4) Grade A

Internships

Applications Development Integration , *Ejada* 2025/08 – 2025/09 | Egypt, Assuit
Gained hands-on experience in application integration and backend development, working with enterprise integration technologies and developing practical software solutions in a professional environment.

Projects

BookStore Web Application , *ASP.NET Core MVC* | *Entity Framework Core* | *SQL Server* | *Identity* (*2FA, RBAC*) | *Bootstrap* 5 2026/06 – 2026/08
Developed a full-stack e-commerce bookstore using MVC and Entity Framework Core, featuring customer browsing, cart, checkout, an admin dashboard for catalog and order management, and secure Identity authentication with 2FA and email confirmation.

GuardianAI — Real-Time Child Safety Mobile Application (Graduation Project) 2025/08 – 2026/06
AI | *Microservices* | *Real-Time Detection* | *Computer Vision* | *NLP* | *YOLOv8* | *Whisper* | *ToxicBERT* | *WebSockets*
Developed a real-time child safety system that detects inappropriate content, violence, and toxic speech using AI-powered detection models. Built independent microservices for nudity detection, violence detection, and toxic speech analysis, enabling real-time monitoring and response. Integrated Computer Vision and NLP models to analyze visual and audio content and identify potentially harmful interactions.
My Role: Backend development, microservices implementation, AI service integration, and real-time communication.

Colorectal Cancer Prediction , *Flask* | *Machine Learning* (*KNN, Naive Bayes, Ensemble*) | *Docker* | *PostgreSQL* | *Flask-Babel* 2026/05 – 2026/06
combining KNN and Naive Bayes models through an Ensemble approach for improved accuracy. Integrated PostgreSQL for data persistence, containerized the app with Docker, and implemented multilingual support (English, Arabic RTL, French).

Virtual_Bank_System , *Java Spring Boot, Microservices, WSO2 API Gateway, Kafka, Docker* 2025/08 – 2025/09
Provides user authentication, account and transaction management, a Backend-for-Frontend (BFF), centralized API management through WSO2 API Gateway, Kafka-based logging, and containerized deployment using Docker.

HealthyBytes , *Flask* (*Python*) | *Recipe Management* | *Nutrition Info* 2025/04 – 2025/05
Developed a Flask web application for exploring categorized recipes with detailed nutritional information to promote healthy eating.

ToDoList , *Flask* (*Python*) | *Docker* 2025/04 – 2025/05
Built a Flask-based task management app with create, read, and update features, containerized using Docker for easy deployment.

Udemy Courses Analysis , *Python* (*Jupyter, Pandas, NumPy, Matplotlib, Scikit-learn*). 2024/08 – 2024/09
Final project in NTI internship: Analyzed datasets using Python, performed exploratory data analysis (EDA), and presented insights with visualizations.

Skills

Programming: Python, Java, C++, C, SQL

Machine Learning & AI: Scikit-learn, Pandas, NumPy, Data Preprocessing, Model Training & Evaluation

Applications Integration: Spring Integration, REST APIs.

Soft Skills: Teamwork, Problem Solving, Communication

Data Analysis: Excel, Power BI, Tableau, Python (Pandas, Matplotlib, Seaborn)

Frameworks & Tools: Spring Boot, Flask, Git, Jupyter Notebook, VS Code and Docker.

Architecture: MVC (Model–View–Controller)

Courses & Certificates

AI & Data Science, *Digital Egypt Pioneers Initiative* (*DEPI*) 2026/06 – Present | Egypt, Assuit

Data Analysis and skills for professional Freelancing , *itida* (*NTI*) 2024/08 – 2024/09 | Remote
Completed a 4-week (120-hour) intensive training program covering data analysis skills tailored for freelance professionals.

Activities

Data Analysis Member – GDG On Campus, Assiut University
contributed to data analysis activities and workshops within a Google-affiliated student developer community.

ECPC for Problem Solving
Participated in the 2024 ICPC ECPC Qualifications, a competitive programming contest under the International Collegiate Programming Contest (ICPC), representing Assiut University.

Languages

- Arabic: Native
- English: Intermediate